# Division properties of exponents and the zero exponent

ANSWER KEY

## Core Concepts: Division properties of exponents and the zero exponent

**1** For the model $f(x) = 3x + 9$ in division properties of exponents and the zero exponent, evaluate $f(2)$.

**a.** $15$ ✓

**b.** $18$

**c.** $6$

**d.** $19$

Substitute $x = 2$: $f(2) = 3(2) + 9 = 15$.

**2** Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

**a.** $x = 1$ and $x = 5$

**b.** $x = 1$ **and** $x = 4$ ✓

**c.** $x = -1$ and $x = -4$

**d.** $x = 5$ and $x = 4$

Factoring $(x - 1)(x - 4) = 0$ yields the roots $x = 1$ and $x = 4$.

**3** A data set recorded for division properties of exponents and the zero exponent consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

**a.** $19.5$

**b.** $20.5$

**c.** $21.5$ ✓

**d.** $23.5$

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by $4$ gives $21.5$.

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## Procedural Fluency & Algebraic Manipulation

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**4** For the model $f(x) = 3x + 9$ in division properties of exponents and the zero exponent, evaluate $f(3)$.

**a.** $18$ ✓

**b.** $21$

**c.** $9$

**d.** $22$

Substitute $x = 3$: $f(3) = 3(3) + 9 = 18$.

**5** Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

**a.** $x = 2$ and $x = 6$

**b.** $x = 2$ **and** $x = 5$ ✓

**c.** $x = -2$ and $x = -5$

**d.** $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

**6** A data set recorded for division properties of exponents and the zero exponent consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

**a.** $19.5$

**b.** $20.5$

**c.** $21.5$ ✓

**d.** $23.5$

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by $4$ gives $21.5$.

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## Graphical & Geometric Interpretation

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**7** For the model $f(x) = 3x + 9$ in division properties of exponents and the zero exponent, evaluate $f(4)$.

**a.** $21$ ✓

**b.** $24$

**c.** $12$

**d.** $25$

Substitute $x = 4$: $f(4) = 3(4) + 9 = 21$.

**8** Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

**a.** $x = 3$ and $x = 7$

**b.** $x = 3$ **and** $x = 6$ ✓

**c.** $x = -3$ and $x = -6$

**d.** $x = 9$ and $x = 18$

Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

**9** A data set recorded for division properties of exponents and the zero exponent consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

**a.** $19.5$

**b.** $20.5$

**c.** $21.5$ ✓

**d.** $23.5$

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by $4$ gives $21.5$.

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## Exam-Style Problem Solving & Modeling

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**10** For the model $f(x) = 3x + 9$ in division properties of exponents and the zero exponent, evaluate $f(5)$.

**a.** $24$ ✓

**b.** $27$

**c.** $15$

**d.** $28$

Substitute $x = 5$: $f(5) = 3(5) + 9 = 24$.

**11** Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

**a.** $x = 4$ and $x = 8$

**b.** $x = 4$ **and** $x = 7$ ✓

**c.** $x = -4$ and $x = -7$

**d.** $x = 11$ and $x = 28$

Factoring $(x - 4)(x - 7) = 0$ yields the roots $x = 4$ and $x = 7$.

**12** A data set recorded for division properties of exponents and the zero exponent consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

**a.** $19.5$

**b.** $20.5$

**c.** $21.5$ ✓

**d.** $23.5$

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by $4$ gives $21.5$.

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